

Code explanation (Call Nr. 9)

Feedback from Call 8: sign conventions, restrictiveness of r_{ij} bound, 3D extension,
comparison with Kurganov
Numerical experiments to compare schemes

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Comment on the document

This document addresses four points raised during last week's call:

- The implicit-step matrix in the document is written as $I - \Delta t L$, while the more common convention is $I + \Delta t L$ with L positive semi-definite (PSD). I clarify what I did in Section 1.
- In Section 2 I comment on how restrictive equation (50) of last week's document is (the local r_{ij} bound on the non-uniform grid), and which growth factors α are even admissible.
- In Section 3 I develop the mathematical reasoning for a 3D extension of the diagonal scheme, in the same style as the 2D derivation.
- In Section 4 I compare our scheme to the Kurganov(-Tadmor) central scheme mathematically; the code-side comparison will be done separately. The focus here is on what each method can and cannot handle.
- In Section 5 I try to numerically show why the new scheme is superior to Kurganov with some small successes, but nothing too crazy unfortunately.
- In Section 6 I run the diagonal solver on slightly longer time-horizons for complicated setups and show that the scheme works really well.
- In Section 7 I present the open todos, for the next weeks.

1 Sign convention: $I \pm \Delta t L$ and PSD vs. NSD

In last week's document it was noted that there might be a sign error. To clarify what I have been doing, I explain this here in more detail.

Recap from last week (orange).

The 2D Fokker–Planck equation is

$$\partial_t U = -\partial_x(F_x U) - \partial_y(F_y U) + \partial_{xx}(D_{xx} U) + \partial_{xy}(D_{xy} U) + \partial_{yy}(D_{yy} U). \quad (1)$$

The diffusion tensor is $\mathbf{D} = \begin{pmatrix} D_{xx} & D_{xy}/2 \\ D_{xy}/2 & D_{yy} \end{pmatrix}$, with entries built from $d_\mu := \beta_\mu C_\mu + \alpha_\mu \geq 0$ and shown to be positive semi-definite by the sum-of-squares argument

$$\mathbf{v}^\top \mathbf{D} \mathbf{v} = \frac{1}{2} \sum_{\mu} d_\mu (m v_x + q v_y)^2 \geq 0 \quad \forall \mathbf{v} \in \mathbb{R}^2. \quad (2)$$

The implicit step in the IMEX time-stepping was written as

$$(I - \Delta t L_h^{\mathcal{D}}) U^{n+1} = U^n + \Delta t L_h^A U^n, \quad (3)$$

and the M-matrix discussion required the off-diagonal entries of $A := I - \Delta t L_h^{\mathcal{D}}$ to be non-positive.

1.1 The two sign conventions

Different approaches lead to $I - \Delta t L$ or $I + \Delta t L$ for the backward Euler step. I explicitly write out both so it is unambiguous which one I have been using.

Convention A (used throughout last week's document). Write the PDE as

$$\partial_t U = \mathcal{L} U, \quad (4)$$

where $\mathcal{L}$ collects all spatial derivatives on the right-hand side of (1). For constant coefficients its Fourier symbol is

$$\widehat{\mathcal{L}^{\mathcal{D}}}(\mathbf{k}) = -(D_{xx} k_x^2 + D_{xy} k_x k_y + D_{yy} k_y^2) = -\mathbf{k}^\top \mathbf{D} \mathbf{k} \leq 0, \quad (5)$$

where the inequality uses precisely the PSD property (2) of $\mathbf{D}$. So the continuous operator $\mathcal{L}^{\mathcal{D}}$ is negative semi-definite (NSD).

The discrete operator $L_h^{\mathcal{D}}$ inherits this property: as we showed last week, its symbol is

$$\widehat{L_h^{\mathcal{D}}}(\xi, \eta) = -\frac{4\tilde{A}_x}{\Delta x^2} \sin^2 \frac{\xi}{2} - \frac{4\tilde{A}_y}{\Delta y^2} \sin^2 \frac{\eta}{2} - \frac{4\tilde{D}_2}{\Delta x \Delta y} \sin^2 \frac{\xi + \sigma \eta}{2} \leq 0 \quad (6)$$

under the non-negativity assumption $\tilde{A}_x, \tilde{A}_y, \tilde{D}_2 \geq 0$. The matrix $L_h^{\mathcal{D}}$ is therefore NSD.

The backward-Euler time step implemented in the code is

$$U^{n+1} = U^n + \Delta t L_h^{\mathcal{D}} U^{n+1} \iff (I - \Delta t L_h^{\mathcal{D}}) U^{n+1} = U^n. \quad (7)$$

Convention B (what would be the more typical). Write the PDE as

$$\partial_t U + A U = 0, \quad (8)$$

where $A = -\mathcal{L}^{\mathcal{D}}$. By construction, A has non-negative spectrum, thus it is PSD as an operator. Concretely, $A = -\partial_{xx}(D_{xx} \bullet) - \partial_{xy}(D_{xy} \bullet) - \partial_{yy}(D_{yy} \bullet)$ in the continuous case, and analogously $A_h = -L_h^{\mathcal{D}}$ in the discrete case, with symbol

$$\widehat{A_h}(\xi, \eta) = -\widehat{L_h^{\mathcal{D}}}(\xi, \eta) \geq 0. \quad (9)$$

Backward Euler then reads

$$U^{n+1} = U^n - \Delta t A_h U^{n+1} \iff (I + \Delta t A_h) U^{n+1} = U^n. \quad (10)$$

This is the form mentioned in the call. The operator A_h here is PSD, and the matrix $I + \Delta t A_h$ has eigenvalues ≥ 1 .

2 Restrictiveness of the local r_{ij} bound (eq. 50 of last week)

Recap from last week (orange).

On a non-uniform grid with cell-local mesh-size ratio

$$\tilde{r}_{ij} = \frac{h_x^+ + h_x^-}{h_y^+ + h_y^-},$$

the splitting takes the same form as on a uniform grid with r replaced by $\tilde{r}_{ij}$:

$$\tilde{D}_{2,ij} = \frac{D_{xy,ij}}{2}, \quad \tilde{A}_{x,ij} = D_{xx,ij} - \tilde{r}_{ij} \tilde{D}_{2,ij}, \quad \tilde{A}_{y,ij} = D_{yy,ij} - \tilde{D}_{2,ij} / \tilde{r}_{ij}. \quad (11)$$

For non-negativity of all three coefficients in every cell we derived

$$\frac{|D_{xy,ij}|}{2 D_{yy,ij}} \leq \tilde{r}_{ij} \leq \frac{2 D_{xx,ij}}{|D_{xy,ij}|} \quad \forall (i, j). \quad (12)$$

This is exactly equation (50) of the call-8 document.

For the geometric grid $\Delta x_{n+1} = (1 + \alpha) \Delta x_n$, $\Delta y_{m+1} = (1 + \alpha) \Delta y_m$ with the same α in both directions,

$$\tilde{r}_{ij} = \frac{\Delta x_i}{\Delta y_j} = \frac{\Delta x_0}{\Delta y_0} (1 + \alpha)^{i-j}. \quad (13)$$

For our model $\mathcal{M} = \{(1, 0), (1, 1)\}$ the bounds reduce to

$$1 \leq \tilde{r}_{ij} \leq 1 + \frac{d_1[i]}{d_2[i]}. \quad (14)$$

Note: Equation 50 is here (12).

2.1 What needs to hold for every cell

The bound (12) is a cell-by-cell constraint. On a geometric grid (13), the ratio $\tilde{r}_{ij}$ varies over $i - j \in \{-(N_y - 1), \dots, (N_x - 1)\}$, so $\tilde{r}_{ij}$ can take values across the range

$$\tilde{r}_{\min} = \frac{\Delta x_0}{\Delta y_0} (1 + \alpha)^{-(N_y - 1)}, \quad \tilde{r}_{\max} = \frac{\Delta x_0}{\Delta y_0} (1 + \alpha)^{N_x - 1}. \quad (15)$$

The dynamic range of $\tilde{r}_{ij}$ is the ratio

$$\frac{\tilde{r}_{\max}}{\tilde{r}_{\min}} = (1 + \alpha)^{(N_x - 1) + (N_y - 1)} = (1 + \alpha)^{N_x + N_y - 2}. \quad (16)$$

This dynamic range is set entirely by α and the grid sizes N_x, N_y .

For (12) to hold simultaneously at every cell, we need the entire interval $[\tilde{r}_{\min}, \tilde{r}_{\max}]$ to fit inside the admissible interval. Take the worst-case cell-local bounds

$$L := \max_{(i,j)} \frac{|D_{xy,ij}|}{2 D_{yy,ij}}, \quad U := \min_{(i,j)} \frac{2 D_{xx,ij}}{|D_{xy,ij}|}, \quad (17)$$

The simultaneous constraint becomes

$$L \leq \tilde{r}_{\min} \quad \text{and} \quad \tilde{r}_{\max} \leq U. \quad (18)$$

Multiplying these two and using (16),

$$\boxed{(1 + \alpha)^{N_x + N_y - 2} = \frac{\tilde{r}_{\max}}{\tilde{r}_{\min}} \leq \frac{U}{L}.} \quad (19)$$

α is bounded purely by the ratio U/L of admissible $\tilde{r}$ -values, and by the linear grid size $N_x + N_y$. Solving for α :

$$\boxed{\alpha \leq \left(\frac{U}{L}\right)^{1/(N_x + N_y - 2)} - 1.} \quad (20)$$

2.2 Plugging in our model

For $\mathcal{M} = \{(1, 0), (1, 1)\}$, recall $D_{xx} = (d_1 + d_2)/2$, $D_{yy} = d_2/2$, $D_{xy} = d_2$, hence

$$\frac{|D_{xy,ij}|}{2D_{yy,ij}} = \frac{d_2[i]}{d_2[i]} = 1, \quad \frac{2D_{xx,ij}}{|D_{xy,ij}|} = \frac{d_1[i] + d_2[i]}{d_2[i]} = 1 + \frac{d_1[i]}{d_2[i]}.$$

So $L = 1$ exactly, and

$$U = \min_i \left(1 + \frac{d_1[i]}{d_2[i]} \right) = 1 + \min_i \frac{d_1[i]}{d_2[i]}.$$

Define $\rho := \min_i d_1[i]/d_2[i]$. Then $U/L = 1 + \rho$, and (20) becomes

$$\boxed{\alpha \leq (1 + \rho)^{1/(N_x + N_y - 2)} - 1 \quad \text{with} \quad \rho = \min_i \frac{d_1[i]}{d_2[i]}.} \quad (21)$$

2.3 What this means in practice

For experiments, we will probably only be allowed to use very small α . One could try to use other meshes, but as shown last week, our method already breaks down if the grid is not geometric. So this does not seem too promising.

In the call, we also discussed ideas of Quadrees or Octrees, but this seems very involved and renders the problem a really non-trivial implementation. This could be subject for further research.

3 Three-dimensional extension of the diagonal scheme

Recap of the 2D scheme (orange).

In 2D the discrete diffusion operator has three pieces: two axial 5-point Laplacians and one diagonal cross stencil $\mathcal{T}_h$ along the D_{xy} -aligned diagonal. The splitting reads

$$\tilde{D}_2 = \frac{D_{xy}}{2}, \quad \tilde{A}_x = D_{xx} - r\tilde{D}_2, \quad \tilde{A}_y = D_{yy} - \tilde{D}_2/r, \quad (22)$$

with $r = \Delta x/\Delta y$. Non-negativity of all three coefficients gives $|D_{xy}|/(2D_{yy}) \leq r \leq 2D_{xx}/|D_{xy}|$, which is non-empty exactly when $\mathbf{D}$ is PSD. The resulting matrix has the M-matrix property and the implicit step preserves positivity.

3.1 The 3D Fokker–Planck equation

In three dimensions we have variables $\mathbf{x} = (x_1, x_2, x_3)$ and a mobile set $\mathcal{M}$ whose elements are triples $\mu = (m_1^{(\mu)}, m_2^{(\mu)}, m_3^{(\mu)}) \in \mathbb{Z}^3$. The Fokker–Planck equation reads

$$\partial_t U = - \sum_{a=1}^3 \partial_{x_a} (F_a U) + \sum_{a=1}^3 \partial_{x_a x_a} (D_{aa} U) + \sum_{1 \leq a < b \leq 3} \partial_{x_a x_b} (D_{ab} U), \quad (23)$$

with drift and diffusion built from the same $d_\mu = \beta_\mu C_\mu + \alpha_\mu \geq 0$ and $\gamma_\mu = \beta_\mu C_\mu - \alpha_\mu$ as in 2D:

$$F_a = \sum_{\mu \in \mathcal{M}} m_a^{(\mu)} \gamma_\mu, \quad a = 1, 2, 3, \quad (24)$$

$$D_{aa} = \frac{1}{2} \sum_{\mu \in \mathcal{M}} (m_a^{(\mu)})^2 d_\mu, \quad a = 1, 2, 3, \quad (25)$$

$$D_{ab} = \sum_{\mu \in \mathcal{M}} m_a^{(\mu)} m_b^{(\mu)} d_\mu, \quad 1 \leq a < b \leq 3. \quad (26)$$

We adopt the convention $D_{ba} := D_{ab}$ for $a < b$ (the diffusion tensor is symmetric).

The 3D diffusion tensor. The associated 3×3 matrix is

$$\mathbf{D} = \begin{pmatrix} D_{11} & D_{12}/2 & D_{13}/2 \\ D_{12}/2 & D_{22} & D_{23}/2 \\ D_{13}/2 & D_{23}/2 & D_{33} \end{pmatrix}. \quad (27)$$

PSD of $\mathbf{D}$ in 3D. For any $\mathbf{v} = (v_1, v_2, v_3)^\top \in \mathbb{R}^3$,

$$\begin{aligned} \mathbf{v}^\top \mathbf{D} \mathbf{v} &= \sum_a D_{aa} v_a^2 + \sum_{a < b} D_{ab} v_a v_b \\ &= \sum_a \left(\frac{1}{2} \sum_\mu (m_a^{(\mu)})^2 d_\mu \right) v_a^2 + \sum_{a < b} \left(\sum_\mu m_a^{(\mu)} m_b^{(\mu)} d_\mu \right) v_a v_b \\ &= \frac{1}{2} \sum_\mu d_\mu \left(\sum_a (m_a^{(\mu)})^2 v_a^2 + 2 \sum_{a < b} m_a^{(\mu)} m_b^{(\mu)} v_a v_b \right) \\ &= \frac{1}{2} \sum_\mu d_\mu \left(\sum_a m_a^{(\mu)} v_a \right)^2 \geq 0, \end{aligned} \quad (28)$$

where the third line collected the diagonal and off-diagonal contributions, and the fourth recognised $\sum_a (m_a^{(\mu)} v_a)^2$ as a complete square. Each summand is non-negative because $d_\mu \geq 0$. Hence $\mathbf{D}$ is PSD in 3D, by the same physical argument as in 2D.

3.2 Three planar diagonal stencils

In 2D we used a single diagonal stencil aligned with the cross-derivative ∂_{xy} . In 3D there are three independent cross-derivative pairs $(1, 2), (1, 3), (2, 3)$, so we introduce three planar diagonal stencils, one in each coordinate plane.

For a uniform grid with spacings Δx_a , $a = 1, 2, 3$, define the unit grid offsets $\mathbf{e}_a$. For each pair $a < b$ define with the same idea again

$$\boxed{\mathcal{T}_h^{(ab)}(U) := \frac{U_{\mathbf{i}+\mathbf{e}_a+\mathbf{e}_b} - 2U_{\mathbf{i}} + U_{\mathbf{i}-\mathbf{e}_a-\mathbf{e}_b}}{\Delta x_a \Delta x_b}}. \quad (29)$$

By Taylor expansion of $U_{\mathbf{i} \pm \mathbf{e}_a \pm \mathbf{e}_b}$ to second order:

$$\begin{aligned} U_{\mathbf{i}+\mathbf{e}_a+\mathbf{e}_b} &= U_{\mathbf{i}} + \Delta x_a \partial_a U + \Delta x_b \partial_b U + \frac{(\Delta x_a)^2}{2} \partial_{aa} U + \Delta x_a \Delta x_b \partial_{ab} U + \frac{(\Delta x_b)^2}{2} \partial_{bb} U + \mathcal{O}(h^3), \\ U_{\mathbf{i}-\mathbf{e}_a-\mathbf{e}_b} &= U_{\mathbf{i}} - \Delta x_a \partial_a U - \Delta x_b \partial_b U + \frac{(\Delta x_a)^2}{2} \partial_{aa} U + \Delta x_a \Delta x_b \partial_{ab} U + \frac{(\Delta x_b)^2}{2} \partial_{bb} U + \mathcal{O}(h^3). \end{aligned}$$

The first-order terms cancel in the sum, and the cross term $\Delta x_a \Delta x_b \partial_{ab} U$ has the same sign in both expansions because both Δx_a and Δx_b flip together. Adding and dividing by $\Delta x_a \Delta x_b$:

$$\mathcal{T}_h^{(ab)}(U) = r_{ab} \partial_{aa} U + 2 \partial_{ab} U + r_{ab}^{-1} \partial_{bb} U + \mathcal{O}(h^2), \quad r_{ab} := \frac{\Delta x_a}{\Delta x_b}. \quad (30)$$

Sign-of- D_{ab} branch. Also here depending on the sign of D_{ab} we could implement a logic to flip the orientations of our stencil like in 2D. I did not further work this out, because we are not planning on implementing this anyways.

To keep notation light, in what follows I assume each $D_{ab} \geq 0$.

3.3 Splitting ansatz

Following the 2D ansatz, we look for non-negative coefficients $\tilde{A}_a$ (for $a = 1, 2, 3$) and $\tilde{D}_{ab}$ (for $a < b$, six unknowns total) such that

$$\boxed{\text{Diff}(C) = \sum_{a=1}^3 \partial_{aa} (\tilde{A}_a C) + \sum_{1 \leq a < b \leq 3} \tilde{D}_{ab} \mathcal{T}_h^{(ab)}(C)}. \quad (31)$$

For matching purposes we replace each $\mathcal{T}_h^{(ab)}$ by its leading-order expansion (30) and require equality with the differential operator

$$\text{Diff}(C) = \sum_a D_{aa} \partial_{aa} C + \sum_{a < b} D_{ab} \partial_{ab} C \quad (\text{constant-coefficient case}). \quad (32)$$

Substituting (30) into (31):

$$\text{Diff}(C) = \sum_a \tilde{A}_a \partial_{aa} C + \sum_{a < b} \tilde{D}_{ab} [r_{ab} \partial_{aa} C + 2 \partial_{ab} C + r_{ab}^{-1} \partial_{bb} C] + \mathcal{O}(h^2). \quad (33)$$

Collect terms by which derivative they contribute to. There are three types of derivative on the right-hand side: ∂_{ab} (cross), ∂_{aa} (axial in direction a), and ∂_{bb} (axial in direction b).

3.3.1 Step 1 : match cross-derivatives

For each pair $a < b$, only the pair stencil $\mathcal{T}_h^{(ab)}$ generates a $\partial_{ab} C$ term in the ansatz; no other piece contains the mixed derivative ∂_{ab} . The pair stencil contributes $2 \tilde{D}_{ab}$ to the coefficient of $\partial_{ab} C$. Matching with the RHS (32) gives, for each $1 \leq a < b \leq 3$:

$$2 \tilde{D}_{ab} = D_{ab} \iff \boxed{\tilde{D}_{ab} = \frac{D_{ab}}{2}, \quad a < b.} \quad (34)$$

3.3.2 Step 2 : collect contributions to each axial second derivative

Now I compute the coefficient of $\partial_{aa}C$ on the right-hand side of (33) for each $a = 1, 2, 3$ separately. Two kinds of terms contribute to $\partial_{aa}C$:

- The dedicated axial term $\partial_{aa}(\tilde{A}_a C)$, contributing $\tilde{A}_a$.
- By (30), the pair stencil $\mathcal{T}_h^{(\alpha, \beta)}$ contributes $r_{\alpha\beta} \partial_{\alpha\alpha}$ and $r_{\alpha\beta}^{-1} \partial_{\beta\beta}$. So:
 - if $\alpha = a$ (and $\beta > a$), the pair contributes $r_{ab} \tilde{D}_{ab}$ to $\partial_{aa}C$ (where $b := \beta$);
 - if $\beta = a$ (and $\alpha < a$), the pair contributes $r_{\alpha\beta}^{-1} \tilde{D}_{\alpha\beta} = r_{ba}^{-1} \tilde{D}_{ba}$ to $\partial_{aa}C$ (where $b := \alpha < a$). Using the convention $r_{ba} := r_{ab}^{-1}$ (which is just $r_{ba} = \Delta x_b / \Delta x_a = (\Delta x_a / \Delta x_b)^{-1} = r_{ab}^{-1}$, a definition not an assumption) and $\tilde{D}_{ba} := \tilde{D}_{ab}$ (which follows from $D_{ba} = D_{ab}$), this contribution can be rewritten as $r_{ab} \tilde{D}_{ab}$.

With these conventions, both cases give the same expression: pair (a, b) with $b \neq a$ contributes $r_{ab} \tilde{D}_{ab}$ to $\partial_{aa}C$, regardless of whether $b > a$ or $b < a$. Total coefficient of $\partial_{aa}C$ on the RHS:

$$\tilde{A}_a + \sum_{b \neq a} r_{ab} \tilde{D}_{ab}. \quad (35)$$

Matching with the LHS coefficient D_{aa} :

$$\tilde{A}_a + \sum_{b \neq a} r_{ab} \tilde{D}_{ab} = D_{aa}, \quad a = 1, 2, 3. \quad (36)$$

To make this concrete, write out the three equations, using the convention $r_{21} = r_{12}^{-1}$ etc.:

$$a = 1 : \quad \tilde{A}_1 + r_{12} \tilde{D}_{12} + r_{13} \tilde{D}_{13} = D_{11}, \quad (37)$$

$$a = 2 : \quad \tilde{A}_2 + r_{12}^{-1} \tilde{D}_{12} + r_{23} \tilde{D}_{23} = D_{22}, \quad (38)$$

$$a = 3 : \quad \tilde{A}_3 + r_{13}^{-1} \tilde{D}_{13} + r_{23}^{-1} \tilde{D}_{23} = D_{33}. \quad (39)$$

3.3.3 Step 3 : substitute $\tilde{D}_{ab} = D_{ab}/2$ from Step 1

Plugging (34) into (37)–(39):

$$\tilde{A}_1 = D_{11} - \frac{1}{2}(r_{12} D_{12} + r_{13} D_{13}), \quad (40)$$

$$\tilde{A}_2 = D_{22} - \frac{1}{2}(r_{12}^{-1} D_{12} + r_{23} D_{23}), \quad (41)$$

$$\tilde{A}_3 = D_{33} - \frac{1}{2}(r_{13}^{-1} D_{13} + r_{23}^{-1} D_{23}). \quad (42)$$

3.3.4 Step 4 : compact form

Each $\tilde{A}_a$ above is D_{aa} minus half a sum over the two pairs involving a , weighted by r_{ab} or $r_{ba} = r_{ab}^{-1}$. Using the conventions $r_{ba} := r_{ab}^{-1}$, $D_{ba} := D_{ab}$ from before, all three formulas (40)–(42) collapse to a single closed form:

$$\boxed{\tilde{D}_{ab} = \frac{D_{ab}}{2} \ (a < b), \quad \tilde{A}_a = D_{aa} - \frac{1}{2} \sum_{b \neq a} r_{ab} D_{ab}, \quad a = 1, 2, 3.} \quad (43)$$

This is the 3D generalisation of the 2D splitting (22).

3.4 Sanity check: the 2D limit

Setting $D_{13} = D_{23} = 0$ in (43):

- $\tilde{D}_{12} = D_{12}/2$, $\tilde{D}_{13} = 0$, $\tilde{D}_{23} = 0$. The pairs (1, 3) and (2, 3) vanish identically.
- $\tilde{A}_3 = D_{33}$ **What do do with this? Can we just ignore it because it goes into a direction that does not even exist in 2D?**
- $\tilde{A}_1 = D_{11} - \frac{1}{2}r_{12}D_{12}$, $\tilde{A}_2 = D_{22} - \frac{1}{2}r_{12}^{-1}D_{12}$.

Identifying $(D_{11}, D_{22}, D_{12}) \leftrightarrow (D_{xx}, D_{yy}, D_{xy})$ and $r_{12} \leftrightarrow r$, this reduces to

$$\tilde{D}_2 = D_{xy}/2, \quad \tilde{A}_x = D_{xx} - r\tilde{D}_2, \quad \tilde{A}_y = D_{yy} - \tilde{D}_2/r,$$

which is exactly the 2D splitting (22).

3.5 Non-negativity in 3D

For the M-matrix property of the implicit step we need each $\tilde{A}_a \geq 0$ and each $\tilde{D}_{ab} \geq 0$. The latter is trivial under the sign convention (or otherwise we would just flip the stencil direction around, as in 2D).

The former gives, for each a ,

$$\boxed{2D_{aa} \geq \sum_{b \neq a} r_{ab} |D_{ab}|, \quad a = 1, 2, 3.} \quad (44)$$

Three linear inequalities in three independent unknowns. The r_{ab} are not independent, by our definition $r_{ab} = \Delta x_a / \Delta x_b$, so

$$r_{12} \cdot r_{23} = r_{13}, \quad r_{12} \cdot r_{23} \cdot r_{31} = 1. \quad (45)$$

So the parameter space reduces to two dimensions. Substituting $r_{13} = r_{12}r_{23}$ into (44) gives a constraint set in (r_{12}, r_{23}) .

3.6 Stencil structure and memory in 3D

The 3D discrete diffusion operator $L_h^{\mathcal{D}, 3D}$ has the following stencil entries per row (i.e. per cell $\mathbf{i}$):

- Self entry at $\mathbf{i}$: $-2(\tilde{A}_1/\Delta x_1^2 + \tilde{A}_2/\Delta x_2^2 + \tilde{A}_3/\Delta x_3^2) - 2(\tilde{D}_{12}/(\Delta x_1\Delta x_2) + \tilde{D}_{13}/(\Delta x_1\Delta x_3) + \tilde{D}_{23}/(\Delta x_2\Delta x_3))$.
- Six axial neighbours at $\mathbf{i} \pm \mathbf{e}_a$ for $a = 1, 2, 3$, each with entry $+\tilde{A}_a/\Delta x_a^2$.
- Six pair-diagonal neighbours at $\mathbf{i} \pm (\mathbf{e}_a + \mathbf{e}_b)$ for the three pairs $a < b$, each with entry $+\tilde{D}_{ab}/(\Delta x_a\Delta x_b)$.

Total: $1 + 6 + 6 = 13$ non-zero entries per row.

For a grid with N_a cells in direction a , the matrix is $N_1N_2N_3$ rows by $N_1N_2N_3$ columns, with $13 N_1N_2N_3$ non-zeros. In the experiments we did so far we used $N \approx 500$, so $500 \cdot 13 \approx 10^9$. With $10^9 \cdot 8\text{bytes} \approx 10^{10}\text{bytes} = 10\text{Gigabytes}$. We have to recompute this matrix every timestep, which suffers from bandwidth constraints, let alone the computational effort involved. This method should therefore not be used in 3D, unless we find a better meshing-option or ways to avoid recomputing all the entries at each timestep.

4 Comparison with the Kurganov(–Tadmor) scheme

Recap of our scheme (orange).

In our IMEX time stepping, the diffusion operator is treated implicitly using the diagonal-stencil splitting derived in last week’s document, while the advection operator $L_h^A = -\partial_x(F_x \bullet) - \partial_y(F_y \bullet)$ is treated explicitly using a fifth-order monotonicity-preserving reconstruction (MP5) combined with a third-order strong-stability-preserving Runge–Kutta time integrator (SSP-RK3). The advective CFL was derived from

$$\Delta t \left(\frac{|F_x|}{\Delta x} + \frac{|F_y|}{\Delta y} \right) \leq \sqrt{3}.$$

On the diffusion side, our innovation is to replace the centred 9-point cross-derivative stencil (which has negative weights at the corners and therefore violates the M-matrix condition when $D_{xy} \neq 0$) by a diagonal-stencil splitting that produces an M-matrix and preserves positivity through the implicit step.

4.1 What the Kurganov–Tadmor paper ([1]) treats

In 2D they address ([1] eq. (4.14))

$$\partial_t u + \partial_x f(u) + \partial_y g(u) = \partial_x Q^x(u, u_x, u_y) + \partial_y Q^y(u, u_x, u_y), \quad (46)$$

where Q^x, Q^y are dissipation fluxes that may depend on both first derivatives u_x and u_y .

The 2D semi-discrete scheme ([1] eq. (4.16)) has the conservative form

$$\frac{d\bar{u}_{j,k}}{dt} = -\frac{H_{j+1/2,k}^x - H_{j-1/2,k}^x}{\Delta x} - \frac{H_{j,k+1/2}^y - H_{j,k-1/2}^y}{\Delta y} + \frac{P_{j+1/2,k}^x - P_{j-1/2,k}^x}{\Delta x} + \frac{P_{j,k+1/2}^y - P_{j,k-1/2}^y}{\Delta y}, \quad (47)$$

where H^x, H^y are the convective fluxes and P^x, P^y are the diffusive fluxes. The convective flux at an x -interface ([1] eq. (4.4)) is

$$H_{j+1/2,k}^x = \frac{f(u_{j+1/2,k}^-) + f(u_{j+1/2,k}^+)}{2} - \frac{a_{j+1/2,k}^x}{2} [u_{j+1/2,k}^+ - u_{j+1/2,k}^-], \quad (48)$$

with $u_{j+1/2,k}^\pm$ the left/right cell-average reconstructions and $a_{j+1/2,k}^x = \max(|f'(u_{j+1/2,k}^-)|, |f'(u_{j+1/2,k}^+)|)$ the local maximum wave speed. The y -flux H^y is the exact analogue with g replacing f . The KT prescription for the diffusive flux at the interface $(j+1/2, k)$ ([1] eq. (4.17)–(4.20)) is the two-point average

$$P_{j+1/2,k}^x = \frac{1}{2} \left[Q^x \left(u_{j,k}, \frac{u_{j+1,k} - u_{j,k}}{\Delta x}, (u_y)_{j,k} \right) + Q^x \left(u_{j+1,k}, \frac{u_{j+1,k} - u_{j,k}}{\Delta x}, (u_y)_{j+1,k} \right) \right], \quad (49)$$

$$P_{j,k+1/2}^y = \frac{1}{2} \left[Q^y \left(u_{j,k}, (u_x)_{j,k}, \frac{u_{j,k+1} - u_{j,k}}{\Delta y} \right) + Q^y \left(u_{j,k+1}, (u_x)_{j,k+1}, \frac{u_{j,k+1} - u_{j,k}}{\Delta y} \right) \right], \quad (50)$$

where the tangential slope $(u_x)_{j,k}, (u_y)_{j,k}$ is a minmod-limited cell-centred value ([1] eq. (5.2)).

4.2 Applying the KT diffusion treatment to our cross-diffusion

4.2.1 Motivation

This whole subsection will try to show, why the scheme from [1] can perform worse than my new solver. **I observed negative values in some Kurganov runs, so I suspected that non-negativity might not be preserved.**

This chapter goes into more implementation details and also tries to hand-wavily show why the scheme might produce negative concentrations. This admittedly needs to be formalized a bit more, but here my knowledge is not sufficient. Broadly speaking I tried to find the faulty 9-point stencil from the beginning inside this scheme, to conclude that the scheme is also not working properly in our case...

4.2.2 Getting explicit formulas

Our diffusion operator is in non-divergence form,

$$\text{Diff}(U) = \partial_{xx}(D_{xx}U) + \partial_{xy}(D_{xy}U) + \partial_{yy}(D_{yy}U),$$

so to use the KT framework I first rewrite it in divergence form. Using $\partial_{xx}(D_{xx}U) = \partial_x[\partial_x(D_{xx}U)]$ and $\partial_{xy}(D_{xy}U) = \frac{1}{2}\partial_x[\partial_y(D_{xy}U)] + \frac{1}{2}\partial_y[\partial_x(D_{xy}U)]$ (**Reasonable?**):

$$\text{Diff}(U) = \underbrace{\partial_x \left[\partial_x(D_{xx}U) + \frac{1}{2}\partial_y(D_{xy}U) \right]}_{=: Q^x} + \underbrace{\partial_y \left[\frac{1}{2}\partial_x(D_{xy}U) + \partial_y(D_{yy}U) \right]}_{=: Q^y}. \quad (51)$$

This works, because the definition of Q allowed us to use u_x, u_y both inside.

Q^x as a function of (u, u_x, u_y) . Treating D_{xx}, D_{xy}, D_{yy} as functions of (x, y) and applying the product rule to the two terms inside Q^x :

$$\begin{aligned} \partial_x(D_{xx}U) &= D_{xx}u_x + (\partial_x D_{xx})u, \\ \frac{1}{2}\partial_y(D_{xy}U) &= \frac{D_{xy}}{2}u_y + \frac{1}{2}(\partial_y D_{xy})u. \end{aligned}$$

Grouping the contributions by which of the three scalar arguments u, u_x, u_y they multiply,

$$\boxed{Q^x(u, u_x, u_y) = D_{xx}u_x + \frac{D_{xy}}{2}u_y + c^x(x, y)u, \quad c^x := \partial_x D_{xx} + \frac{1}{2}\partial_y D_{xy}.} \quad (52)$$

The analogous calculation for Q^y gives $Q^y(u, u_x, u_y) = \frac{D_{xy}}{2}u_x + D_{yy}u_y + c^y u$ with $c^y := \frac{1}{2}\partial_x D_{xy} + \partial_y D_{yy}$.

Assembling $P_{j+1/2, k}^x$. KT's flux (49) evaluates Q^x at the cell-centres on the left and right of the interface $(j+1/2, k)$ and averages. On the left side, cell (j, k) , the three slots of (52) are filled as $(u, u_x, u_y) \leftarrow (u_{j, k}, (u_{j+1, k} - u_{j, k})/\Delta x, (u_y)_{j, k})$, where the u_x -slot uses the interface centred difference and the u_y -slot uses the cell-centred (possibly limited) slope. On the right side, cell $(j+1, k)$, the u_x -slot is the same interface difference, while the u_y -slot becomes $(u_y)_{j+1, k}$. Substituting into (52) on each side and then taking the half-sum:

$$P_{j+1/2, k}^x = \frac{1}{2} \left[D_{xx, j, k} \frac{u_{j+1, k} - u_{j, k}}{\Delta x} + \frac{D_{xy, j, k}}{2} (u_y)_{j, k} + c_{j, k}^x u_{j, k} \right] + \frac{1}{2} \left[D_{xx, j+1, k} \frac{u_{j+1, k} - u_{j, k}}{\Delta x} + \frac{D_{xy, j+1, k}}{2} (u_y)_{j+1, k} + c_{j+1, k}^x u_{j+1, k} \right]. \quad (53)$$

The u_x -slot carries the same interface difference on both sides, so its two D_{xx} prefactors combine into the interface midpoint $\bar{D}_{xx, j+1/2} := (D_{xx, j, k} + D_{xx, j+1, k})/2$. Collecting by piece:

$$P_{j+1/2, k}^x = \underbrace{\bar{D}_{xx, j+1/2} \frac{u_{j+1, k} - u_{j, k}}{\Delta x}}_{\text{axial}} + \underbrace{\frac{1}{4} [D_{xy, j, k} (u_y)_{j, k} + D_{xy, j+1, k} (u_y)_{j+1, k}]}_{\text{cross}} + \underbrace{\frac{1}{2} [c_{j, k}^x u_{j, k} + c_{j+1, k}^x u_{j+1, k}]}_{\text{lower-order}}. \quad (54)$$

Reduction to a centred 9-point stencil in smooth regions. I now take the x -flux divergence $(P_{j+1/2, k}^x - P_{j-1/2, k}^x)/\Delta x$ and track each of the three pieces in (54) separately. Note that $P_{j-1/2, k}^x$ is the flux at the interface between cells $(j-1, k)$ and (j, k) , so it is obtained from (54) by shifting $j \rightarrow j-1$.

4.2.3 Investigation of the components

Axial piece. The axial terms give

$$\frac{1}{\Delta x} \left[\bar{D}_{xx, j+1/2} \frac{u_{j+1, k} - u_{j, k}}{\Delta x} - \bar{D}_{xx, j-1/2} \frac{u_{j, k} - u_{j-1, k}}{\Delta x} \right],$$

the standard variable-coefficient 3-point x -Laplacian. This is identical to the axial discretisation we already use.

Cross piece. Subtracting the cross piece of $P_{j-1/2,k}^x$ from that of $P_{j+1/2,k}^x$:

$$\frac{P_{j+1/2,k}^{x, \text{cross}} - P_{j-1/2,k}^{x, \text{cross}}}{\Delta x} = \frac{1}{\Delta x} \cdot \frac{1}{4} \left[\underbrace{D_{xy,j,k}(u_y)_{j,k}}_{(a)} + D_{xy,j+1,k}(u_y)_{j+1,k} - D_{xy,j-1,k}(u_y)_{j-1,k} - \underbrace{D_{xy,j,k}(u_y)_{j,k}}_{(a)} \right]. \quad (55)$$

The two terms labelled (a) are identical and cancel exactly. What remains is

$$\Pi_{j,k}^{xy} := \frac{1}{4 \Delta x} \left[D_{xy,j+1,k}(u_y)_{j+1,k} - D_{xy,j-1,k}(u_y)_{j-1,k} \right]. \quad (56)$$

In smooth regions where the limiter reduces to the centred difference $(u_y)_{j,k} = (u_{j,k+1} - u_{j,k-1})/(2\Delta y)$, and assuming constant D_{xy} for clarity:

$$\begin{aligned} \Pi_{j,k}^{xy} &= \frac{D_{xy}}{4 \Delta x} \left[\frac{u_{j+1,k+1} - u_{j+1,k-1}}{2\Delta y} - \frac{u_{j-1,k+1} - u_{j-1,k-1}}{2\Delta y} \right] \\ &= \frac{D_{xy}}{8 \Delta x \Delta y} \left[u_{j+1,k+1} - u_{j+1,k-1} - u_{j-1,k+1} + u_{j-1,k-1} \right]. \end{aligned} \quad (57)$$

The symmetric contribution from $(P_{j,k+1/2}^y - P_{j,k-1/2}^y)/\Delta y$, by the identical cancellation argument applied in the y -direction, gives

$$\Pi_{j,k}^{yx} = \frac{D_{xy}}{8 \Delta x \Delta y} \left[u_{j+1,k+1} - u_{j-1,k+1} - u_{j+1,k-1} + u_{j-1,k-1} \right], \quad (58)$$

which is the same expression. Adding both, as done in our original formula (47):

$$\boxed{\Pi_{j,k}^{xy} + \Pi_{j,k}^{yx} = \frac{D_{xy}}{4 \Delta x \Delta y} \left[u_{j+1,k+1} - u_{j+1,k-1} - u_{j-1,k+1} + u_{j-1,k-1} \right]}. \quad (59)$$

This is the standard centred 9-point cross-derivative stencil, which we started with. We exactly wanted to avoid it, because this can produce negative values.

Lower-order piece. The lower-order terms in (54) give

$$\frac{1}{\Delta x} \cdot \frac{1}{2} \left[c_{j,k}^x u_{j,k} + c_{j+1,k}^x u_{j+1,k} - c_{j-1,k}^x u_{j-1,k} - c_{j,k}^x u_{j,k} \right] = \frac{c_{j+1,k}^x u_{j+1,k} - c_{j-1,k}^x u_{j-1,k}}{2 \Delta x},$$

a centred first difference of $c^x u$ in x . This touches only the east and west neighbours $u_{j\pm 1,k}$, never the four corner nodes $u_{j\pm 1,k\pm 1}$.

Regarding its effect on the M-matrix and positivity: to be honest I am not sure we can simply ignore this piece. Its sign at $u_{j+1,k}$ is $+c_{j+1,k}^x/(2\Delta x)$, which can be positive or negative depending on how D_{xx} and D_{xy} vary spatially, so it could in principle produce a positive off-diagonal entry in the east or west direction. However, that would be a separate effect from the corner-weight violation identified in (59), and I don't think they cancel each other out. But in this part I am not sure...

4.2.4 What about the limiter?

Outside the smooth regime the cell-centred slope $(u_y)_{j,k}$ is replaced by a minmod-limited value, so (57) is not exact.

In our problem we often use distributions like smooth Gaussian-like profiles. The minmod limiter is then inactive almost everywhere, and (59) is the operative stencil. So we should be able to see this effect in experiments.

4.3 Stability and time integration in KT vs. in our scheme

There is a second important difference, on the time-integration side. KT integrates the full RHS (convection + diffusion) in eq. (4.16) by an explicit Runge–Kutta scheme (Table 5.1, SSP-RK3). In the parabolic regime this produces a parabolic CFL constraint of the form

$$\Delta t \lesssim \min\left(\frac{\Delta x^2}{\|D\|}, \frac{\Delta y^2}{\|D\|}\right), \quad (60)$$

which becomes severe when the grid is fine. Additionally, the maximum-principle proof (Theorem 5.1) for the convective sub-scheme (KT eq. (5.1)) requires the convective CFL condition (5.3),

$$\max\left(\frac{\Delta t}{\Delta x} \max |f'(u)|, \frac{\Delta t}{\Delta y} \max |g'(u)|\right) \leq \frac{1}{8}, \quad (61)$$

and is stated for the convective part (KT eq. (5.1)) only, not for the full scheme (KT eq. (4.16)).

Our scheme, by contrast, is IMEX: diffusion is implicit, so the parabolic CFL (60) is absent and only the advective CFL $\Delta t(|F_x|/\Delta x + |F_y|/\Delta y) \leq \sqrt{3}$ remains. For long-time integration of our model this is a clear advantage.

4.4 Implementation of the KT scheme

The KT scheme has been implemented with the following components:

- **Advection:** MP5 reconstruction, replacing the MUSCL/minmod reconstruction assumed in [1] (cf. eq. (2.4)–(2.5) there), as proposed by Mr. Jourdan.
- **Diffusion:** KT diffusion fluxes from eq. (4.20) of [1], with minmod-limited cell-centred slopes (eq. (5.2)) for the tangential derivative contributions $(u_y)_{j,k}$ and $(u_x)_{j,k}$ in (54).
- **Time integration:** SSP-RK3 (also proposed in Table 5.1 of [1]). Fully explicit in both advection and diffusion.

4.5 Scope of the KT maximum principle

Theorem 5.1 and Corollary 5.1 of [1] establish a maximum-principle-satisfying property for a pure hyperbolic semi-discrete scheme.

This theorem does not apply to the full convection-diffusion scheme (4.16).

When the diffusion fluxes P^x, P^y from eq. (4.20) are included to form the full semi-discrete scheme (4.16), Kurganov and Tadmor do not prove any maximum principle or positivity result. As roughly shown in Section 4.2, in smooth regions the KT diffusion discretisation based can potentially produce negative values in worst-cases.

5 Comparison plots

Three methods are compared side by side: the **9-pt IMEX** (implicit cross-diffusion, M-matrix violated), **KT** (Kurganov–Tadmor, explicit MP5+SSP-RK3), and the **Diagonal IMEX** (M-matrix, exactly positivity-preserving). All experiments use domain $x \in [2, 90]$, $y \in [2, 36]$, $\Delta x = \Delta y = 0.5$, $\varepsilon = 1$, $\sigma = 0.9$. Row 1 of each comparison shows the solution snapshot at T_{end} (RdBu colormap, $\pm \max U_0$). Row 2 shows the **negative** part $\max(0, -U)$ at T_{end} on a shared Reds colorscale whose maximum equals the worst history min across both non-diagonal methods.

5.1 Case A: Near-singular diffusion tensor, diagonal step

Setup. $C_{10} = 0.1$, $C_{11} = 1$ (fixed), unnormalized diagonal step ($\max U_0 = 1$), $T = 1$. The ratio $C_{11}/C_{10} = 10$ makes the axial diffusion coefficient $A_x = d_1/2 \approx 0.05\sqrt{x}$ much smaller than $D_{xy} = d_2 \approx \sqrt{x}$, pushing the diffusion tensor close to singular.

What we see. The 9-pt IMEX accumulates negatives monotonically to $\min U \approx -1.4 \times 10^{-3}$ at $T = 1$ (persistent red band at the diagonal interface in row 2). KT spikes to -6×10^{-4} at $t \approx 0.02$ then fully recovers to machine-zero by $T = 1$. The diagnostic time plot shows the characteristic shapes: a transient KT spike versus a monotonically growing 9-pt IMEX violation.

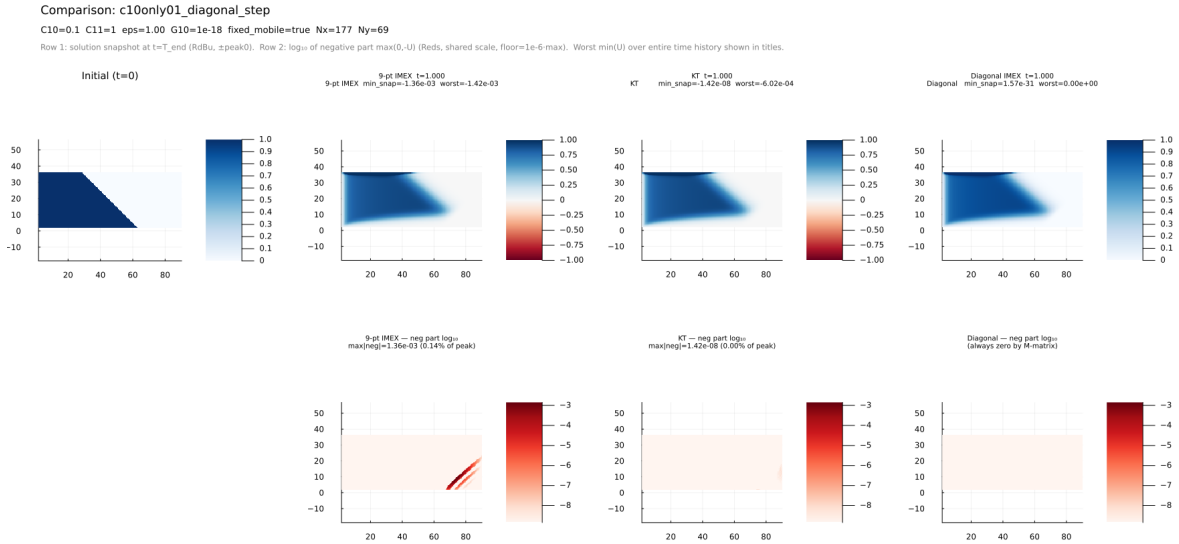


Figure 1: Case A: solution snapshots and negative-part panels at $T = 1$.

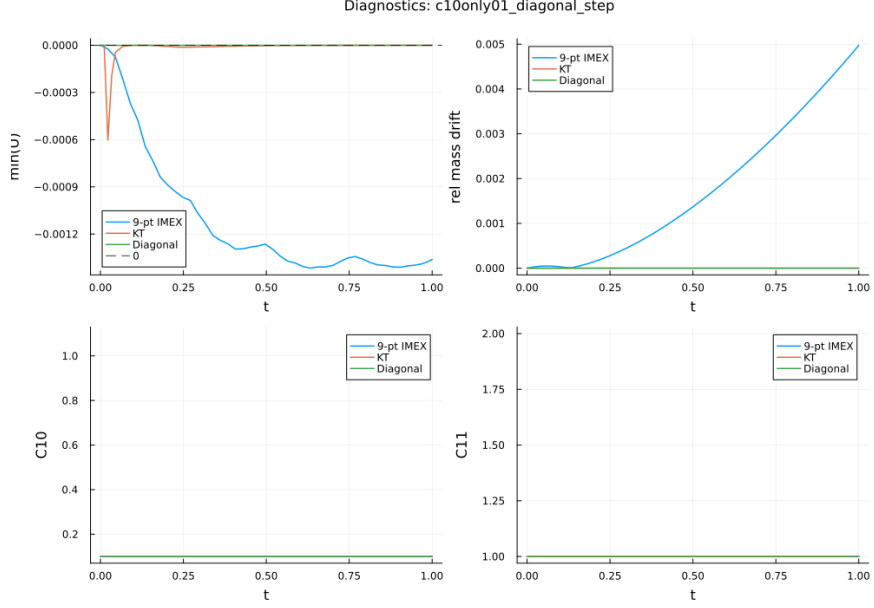


Figure 2: Case A: time histories of $\min(U)$, mass drift, and concentrations.

5.2 Case B: Point mass, near-singular tensor

Setup. $C_{10} = 0.1$, $C_{11} = 1$ (fixed), normalized tight Gaussian ($\sigma = \Delta x$), $T = 2$.

What we see. The 9-pt IMEX violates positivity from the very first step and remains at $\min U \approx -1.25 \times 10^{-3}$ for the entire run. KT stays at machine epsilon throughout (white panel).

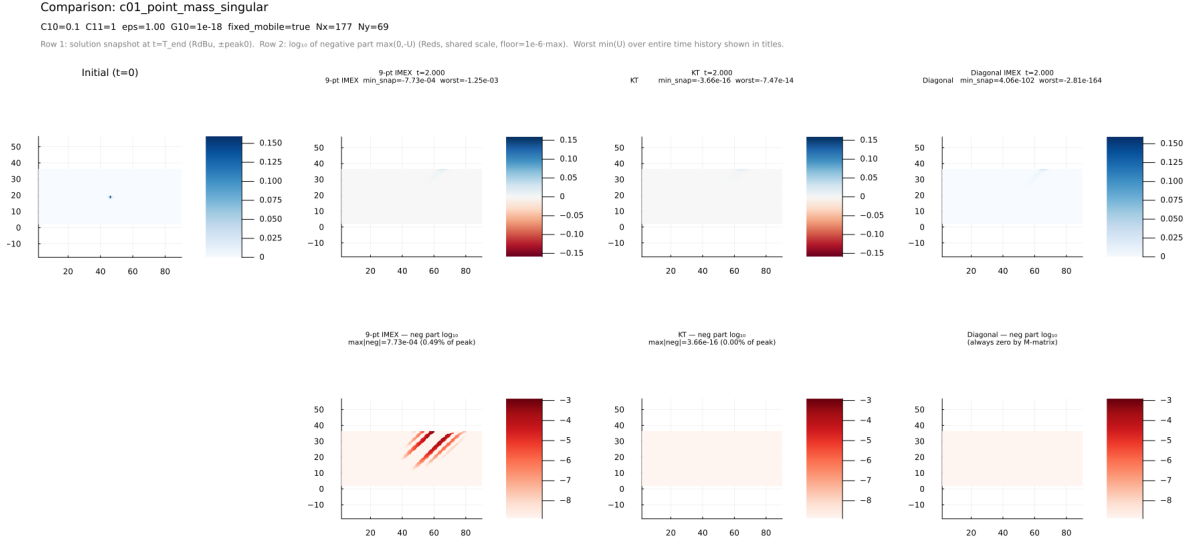


Figure 3: Case B: solution snapshots and negative-part panels at $T = 2$.

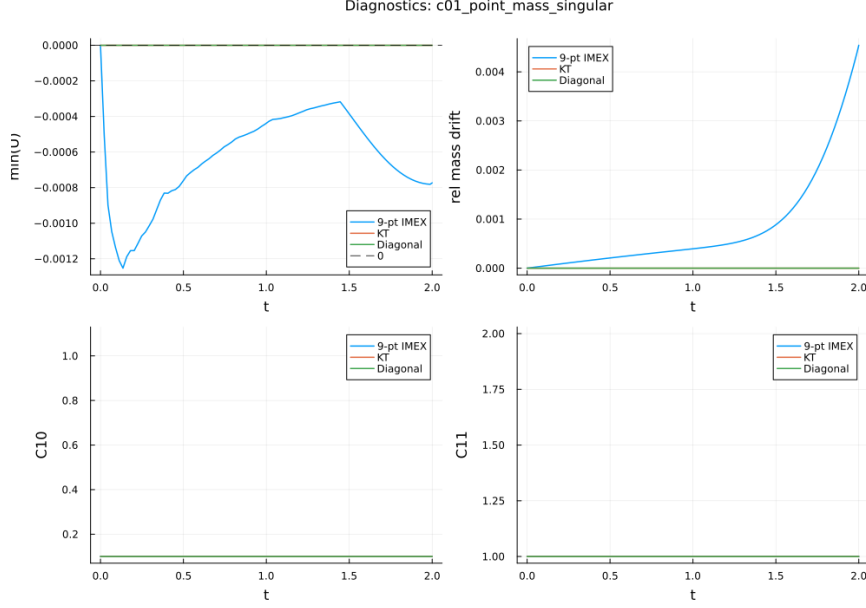


Figure 4: Case B: time histories. Note growing mass drift for 9-pt IMEX.

5.3 Case C: Dynamic concentrations, failure pattern reverses

Setup. $C_{10} = 0.1$, $C_{11} = 1$, unnormalized diagonal step, `fixed_mobile=false`, $T = 2$. The mobile concentrations C_{10} , C_{11} evolve with the solution and decay rapidly to near-zero (the equilibrium reaction consumes the mobile species).

What we see. The failure pattern reverses. The 9-pt IMEX is machine-epsilon clean throughout, while KT develops a persistent violation growing to $\min U \approx -5.3 \times 10^{-4}$ at $T = 2$ (faint red spot at the interface in row 2). **I have the suspicion that the CFL condition failed here for KT, but I will need to check if this is a method failure or just me doing a mistake (it's weird that it happens on border and so abruptly)**

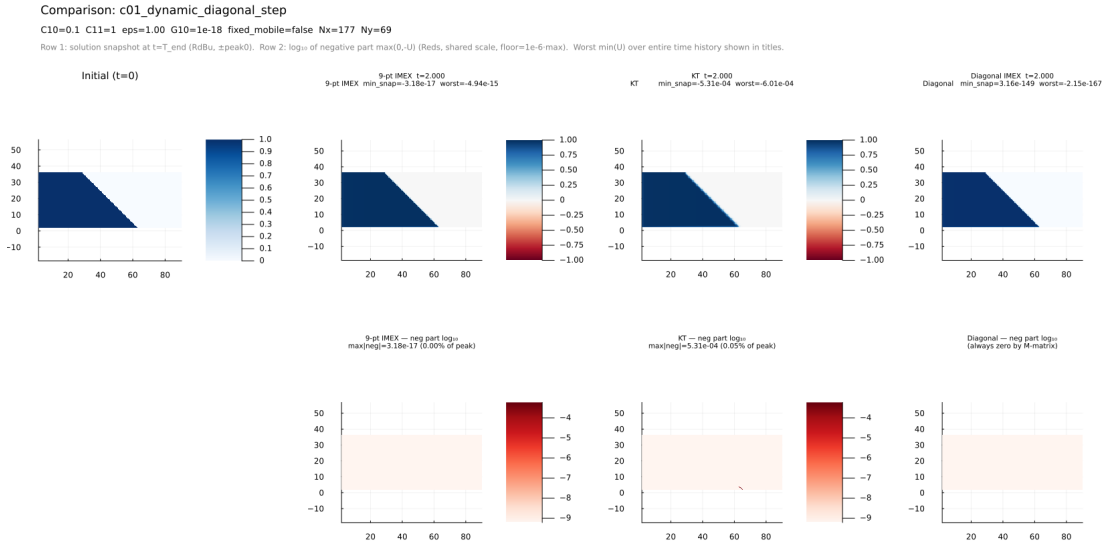


Figure 5: Case C: solution snapshots and negative-part panels at $T = 2$. Almost invisible, but there are some pixels appearing in KT scheme at bottom in the middle.

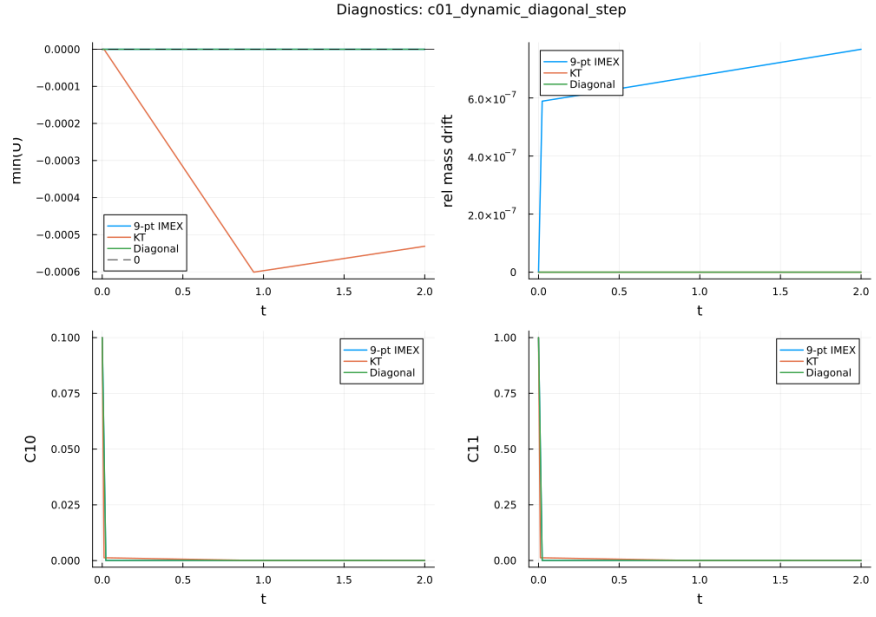


Figure 6: Case C: C_{11} and C_{10} collapse to near-zero within $t \approx 0.1$.

6 Long-run experiments (diagonal solver)

All five cases use only the diagonal IMEX solver, which guarantees $\min U \geq 0$ at every step. The runs demonstrate stable, physically consistent long-time behaviour. Note that the time horizons shown are still relatively modest compared to the full physical dynamics. I still need to test longer runs.

6.1 Case 1: Axis-aligned step $\rightarrow$ quasi-steady state

$C_{10} = 0.3$, $C_{11} = 0.1$, step ($x < 30$), dynamic, $T = 1000$. Mass initially concentrated in the left half diffuses and drifts toward a quasi-steady distribution; the signed-log panels reveal the slowly spreading low-density tails.

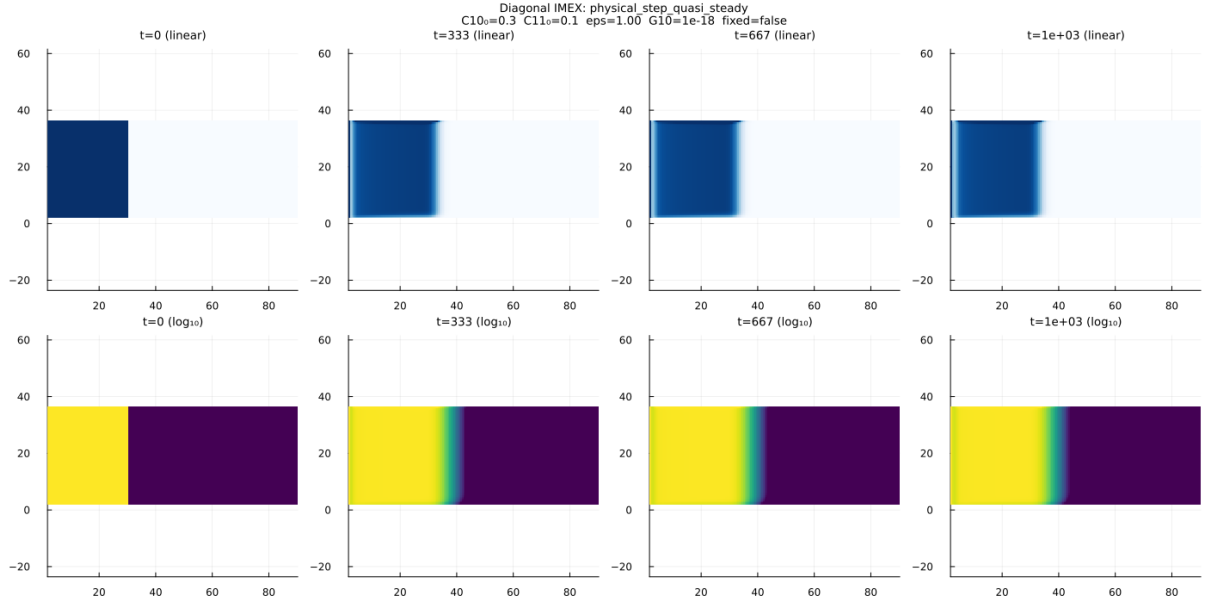


Figure 7: Case 1: four snapshots (linear and $\log_{10}$ colormaps).

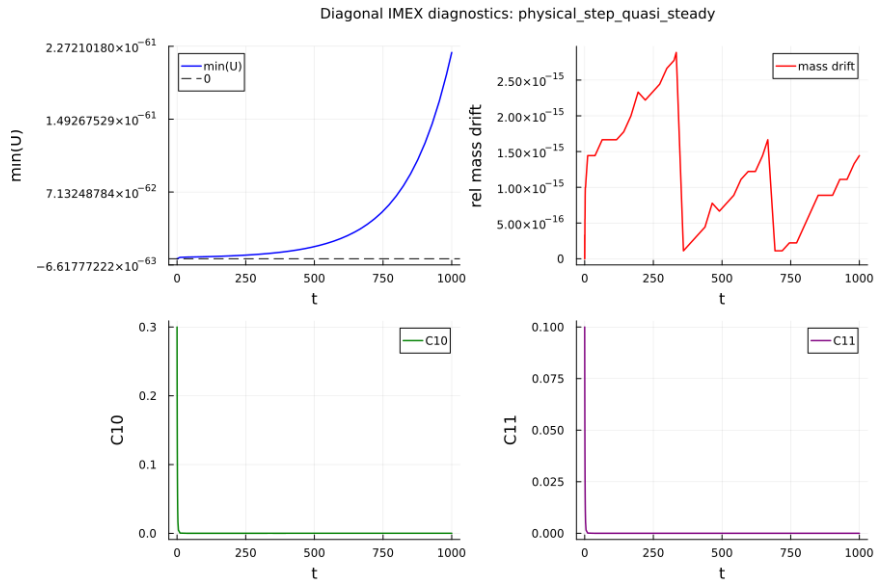


Figure 8: Case 1: time histories of $\min(U)$, mass, and concentrations.

6.2 Case 2: Gaussian blob → anisotropic spreading

$C_{10} = 0.2$, $C_{11} = 0.15$, Gaussian centred at (30, 12), dynamic, $T = 1000$. The positive cross-diffusion $D_{xy} > 0$ elongates the blob along the NE-SW diagonal. As C_{11} decays the diffusion becomes more isotropic, visible in the later snapshots.

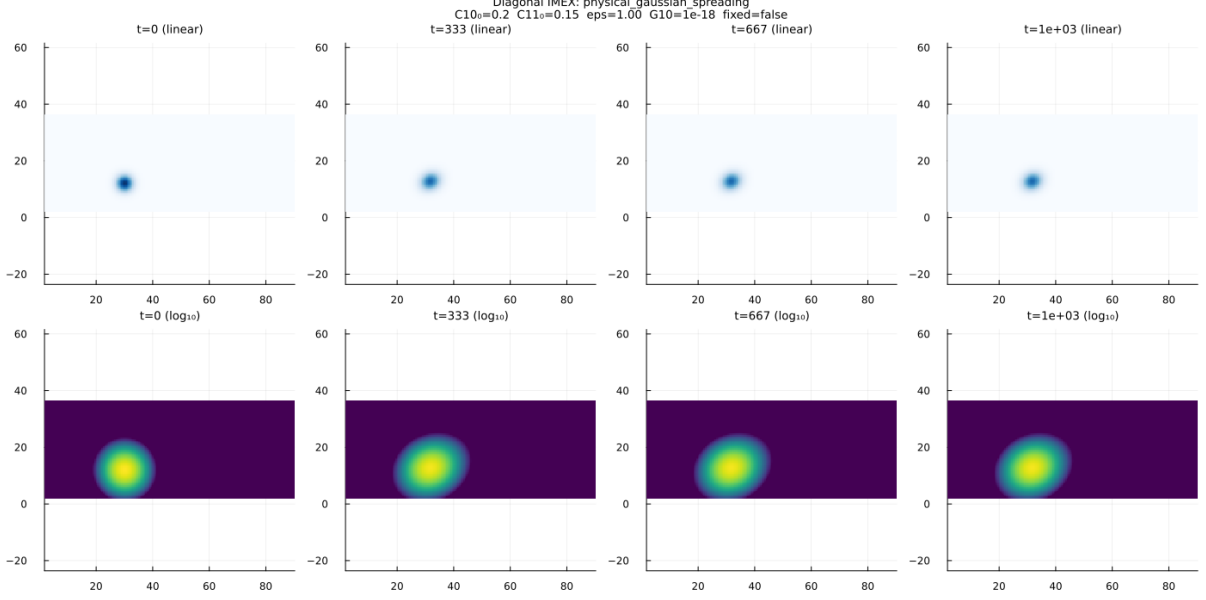


Figure 9: Case 2: anisotropic spreading; elongation along NE-SW visible at early times.

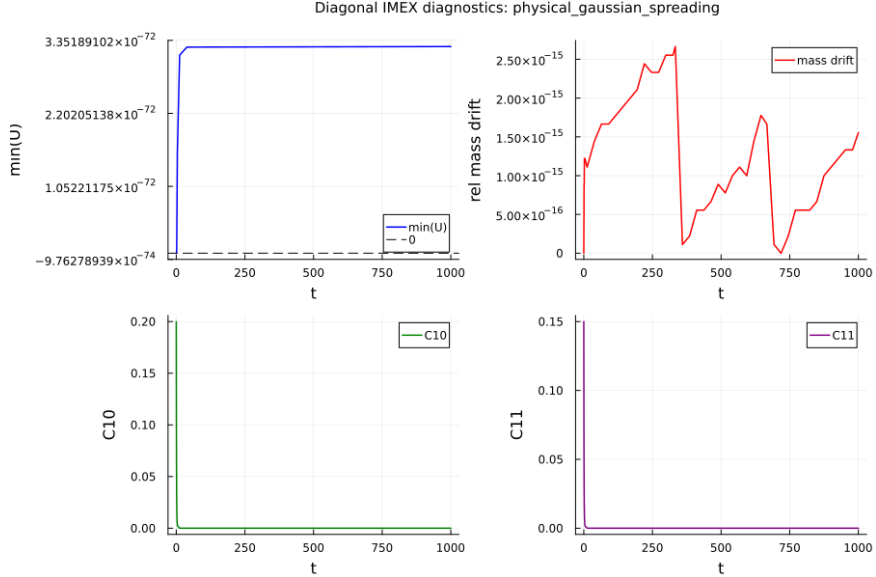


Figure 10: Case 2: diagnostics.

6.3 Case 3: Extreme fixed coupling

$C_{10} = C_{11} = 1$ (fixed throughout), diagonal step, $T = 1000$. This is the maximum cross-diffusion stress: $D_{xy} = d_2 \approx \sqrt{x}$ remains large for the entire run. The diagonal solver runs to $T = 1000$ with $\min U \geq 0$ at every step.

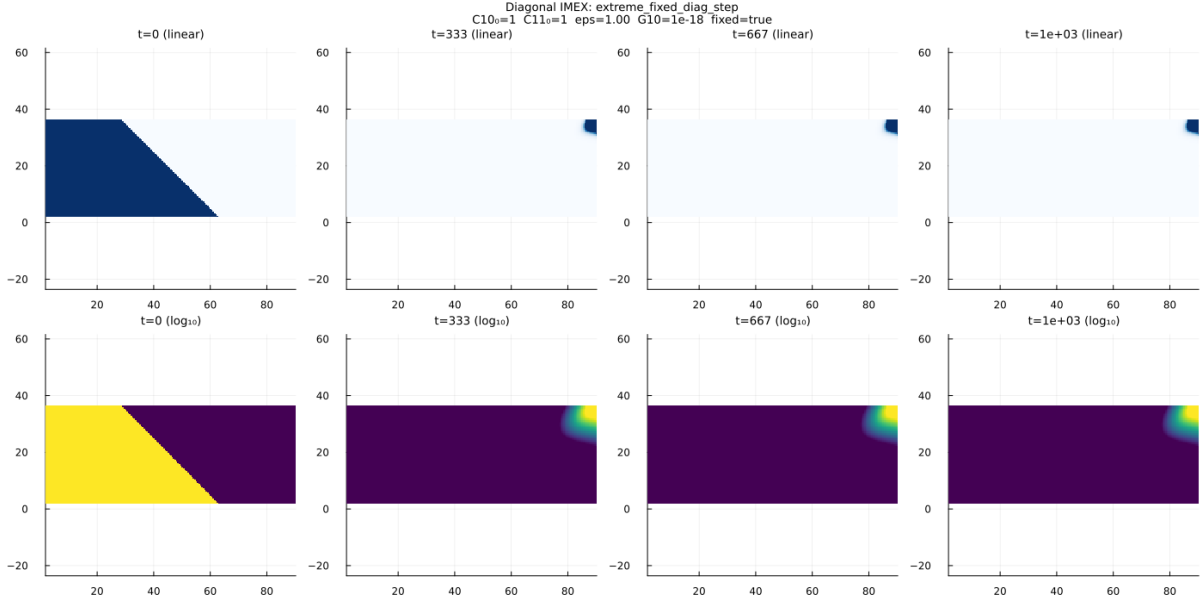


Figure 11: Case 3: extreme fixed coupling; only the diagonal solver remains positive.

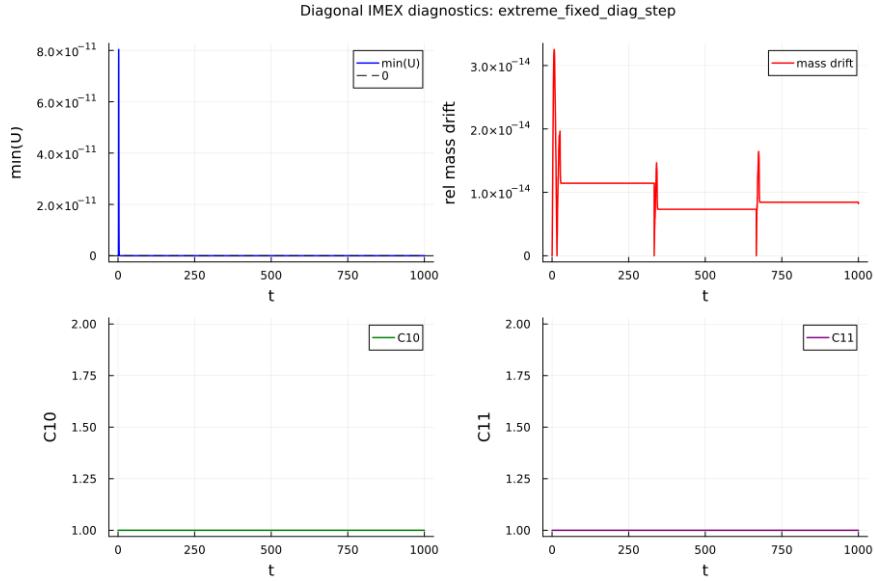


Figure 12: Case 3: diagnostics.

6.4 Case 4: Mass injection from source

$C_{10} = 0.05$, $C_{11} = 0.01$, $G_{10} = 10^{-17}$ (continuous source), nearly uniform IC, $T = 100$. The source continuously injects mass into C_{10} . The images don't show much, but the plots show clearly some movement. I think running this on longer time-horizons could be interesting. But it was mentioned in the call, that without the Bekker-Döring coupling we might not see anything.

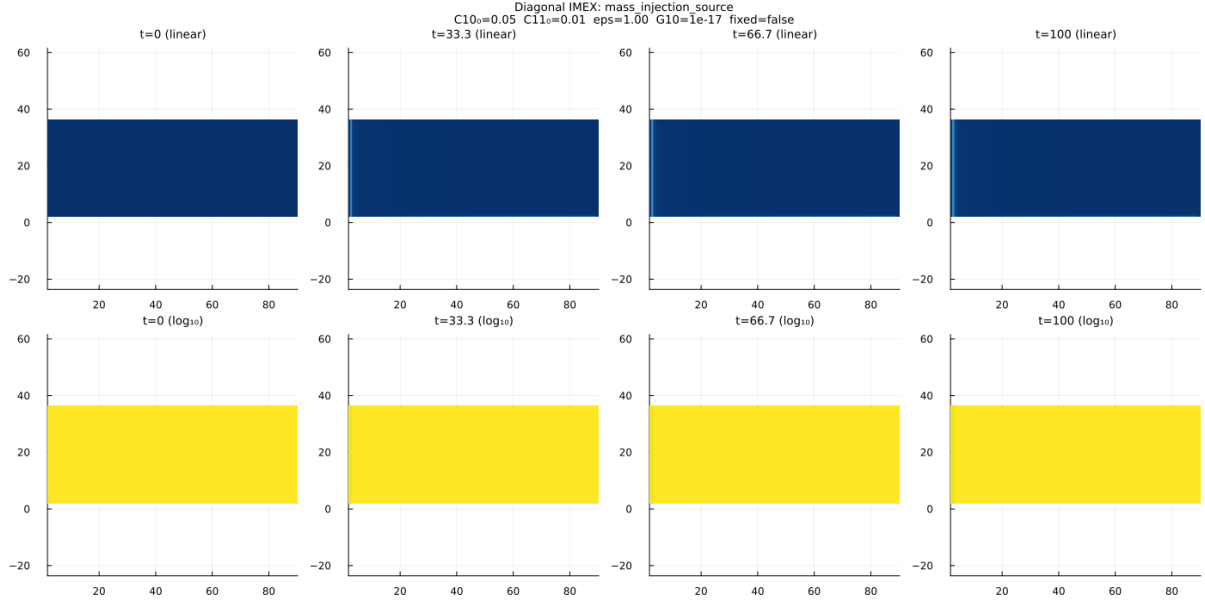


Figure 13: Case 4: mass injection; total mass grows with time.

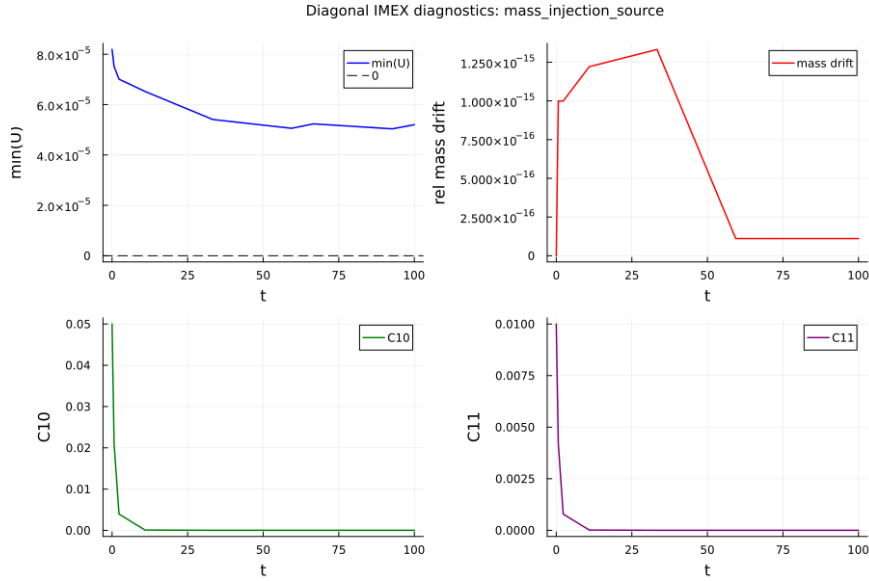


Figure 14: Case 4: diagnostics showing growing total mass.

6.5 Case 5: Concentration-driven dynamics, box IC

$C_{10} = 0.5$, $C_{11} = 0.3$, box ($x < 40$, $y < 14$), dynamic, $T = 1000$. Large initial C_{11} creates strong NE-SW anisotropy. As C_{11} decays the diffusion becomes more isotropic; the change in elongation direction is visible across the four snapshots.

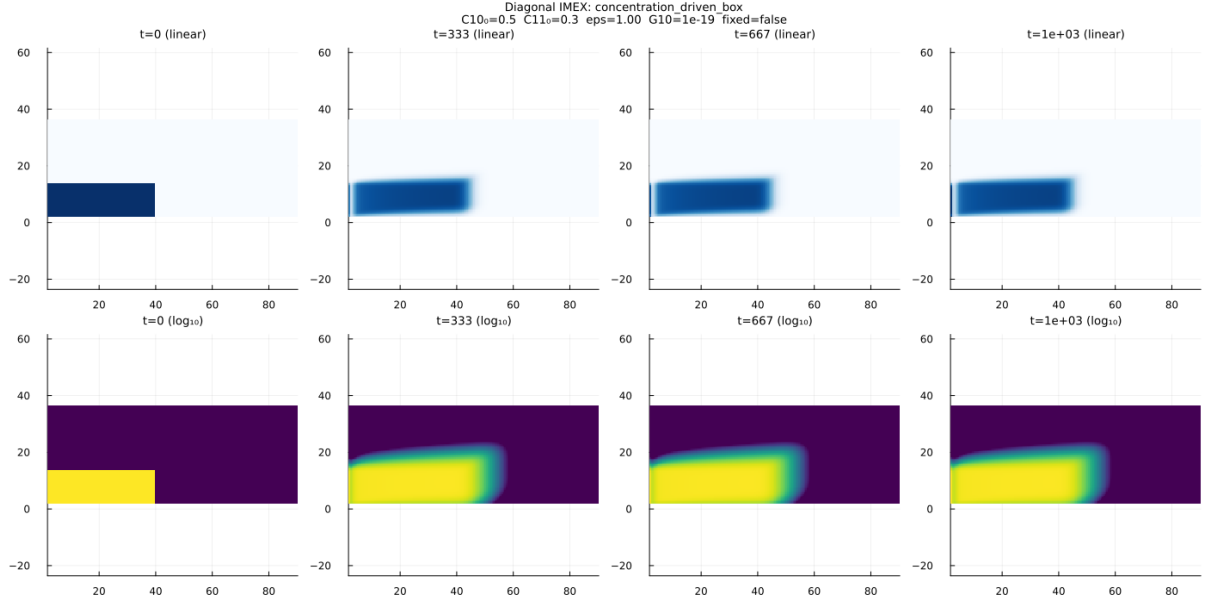


Figure 15: Case 5: concentration-driven dynamics; anisotropy diminishes as $C_{11} \rightarrow 0$.

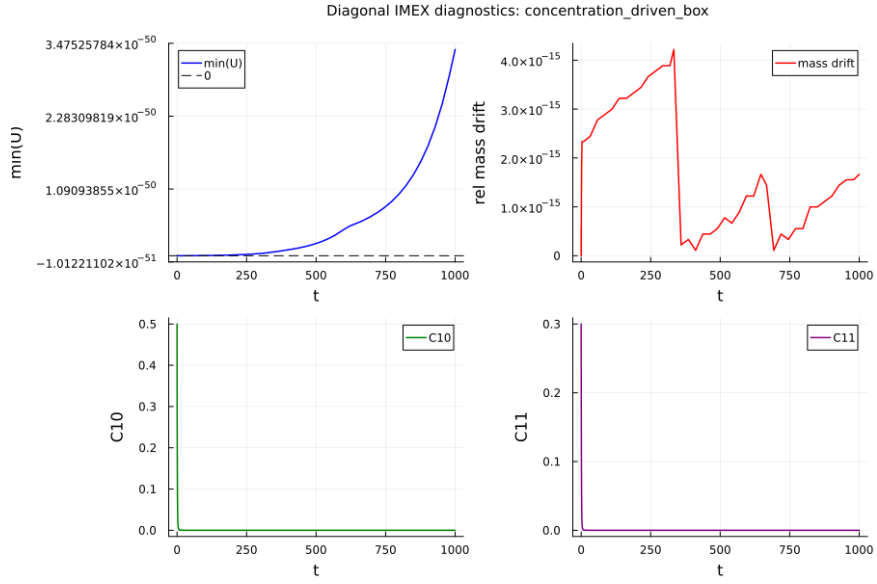


Figure 16: Case 5: diagnostics.

7 Open todos

- Implement the geometric grid for the Diagonal Solver.
- Implement the improved data-output frequency as proposed by Mr. Monasse, based on how much our concentration changes relatively over time.
- Compare Bekker-Döring and the diagonal solver.
- Finish proofreading the code for Kurganov and push to Git.
- Create a new clean folder on the Github repo, with only the code that is actually used.
- Run longer runs with diagonal solver.

References

- [1] Alexander Kurganov and Eitan Tadmor. “New High-Resolution Central Schemes for Non-linear Conservation Laws and Convection–Diffusion Equations”. In: *Journal of Computational Physics* 160.1 (2000), pp. 241–282. ISSN: 0021-9991. DOI: <https://doi.org/10.1006/jcph.2000.6459>. URL: <https://www.sciencedirect.com/science/article/pii/S0021999100964593>.